ITA0605 MACHINE LEARNING

ASSIGNMENT 1

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CO1: Learning problems and applications, Perspectives and issues in machine learning, Types of learning, Concept learning task, Version spaces, Candidate elimination algorithm, Decision tree learning algorithm, Heuristic search in decision trees, Overfitting and underfitting

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Assignment Title: Design, Analyse and Validate an Intelligent Early-Warning DecisionSupport System for Academic Risk Identification Using Concept Learning and Decision
Tree Learning (**Total Marks:** 50)

Duration: 1 Week

Scenario – University Academic Early-Warning System

A university wishes to identify, early in the semester, students who are **at risk of failing** so that the academic office can offer timely counselling, remedial classes or mentoring. Advisors currently flag students manually, which is slow, inconsistent between advisors and not scalable across thousands of students.

The university has a small, labelled history of past students. For each student, the following **categorical attributes** are recorded after week 6 of the semester:

Attribute	Values	Operational meaning
Attendance	Low, Medium, High	Low < 65 %; Medium 65–80 %; High > 80 %
Assignments	Irregular, Regular	Regular = at least 80 % submitted on time
StudyHours	Low, High	Self-reported hours per week: Low < 8 h
PriorCGPA	Weak, Strong	Weak < 6.0; Strong ≥ 6.0

LMS	Passive, Active	Activity on the learningmanagement system
Mentoring	No, Yes	Enrolled in a mentoring programme

The **target concept** is **AtRisk** (Yes / No), assigned by the academic council at the end of the semester.

You are required to investigate, design and compare **concept-learning** and **decisiontree** approaches for this decision-support system, and to determine which approach (or combination) is appropriate. The system must produce **interpretable** decisions, because advisors must be able to explain to a student why he or she was flagged.

Design Challenge

After one semester of use the university reports that the system must now cope with **additional training examples, noisy labels from different advisors, previously unseen attribute combinations and values, and a higher accuracy requirement** (in particular, very few at-risk students may be missed). Re-evaluate your original design and decide whether it should be **retained, modified or redesigned** (see the Day-4 Design Challenge).

Datasets Provided Dataset A – Concept-learning dataset (5 examples, 6 attributes)

#	Attendance	Assignments	StudyHours	PriorCGPA	LMS	Mentoring	AtRisk
E1	Low	Irregular	Low	Weak	Passive	No	Yes
E2	Low	Irregular	Low	Strong	Passive	No	Yes
E3	High	Regular	High	Strong	Active	Yes	No
E4	Low	Irregular	High	Weak	Passive	Yes	Yes
E5	Medium	Irregular	High	Strong	Active	No	No

Dataset B – Decision-tree training dataset (14 examples, 4 attributes)

#	Attendance	Assignments	LMS	PriorCGPA	AtRisk
1	Low	Irregular	Passive	Weak	Yes
2	Low	Irregular	Passive	Strong	Yes

3	Low	Regular	Active	Weak	Yes
4	Low	Regular	Active	Strong	No
5	Medium	Irregular	Passive	Weak	Yes
6	Medium	Irregular	Active	Strong	No
#	Attendance	Assignments	LMS	PriorCGPA	AtRisk
7	Medium	Regular	Passive	Weak	Yes
8	Medium	Regular	Active	Strong	No
9	High	Irregular	Passive	Weak	Yes
10	High	Irregular	Active	Strong	No
11	High	Regular	Passive	Strong	No
12	High	Regular	Active	Weak	No
13	Medium	Irregular	Passive	Strong	No
14	Low	Irregular	Active	Weak	Yes

Test Set T – held-out students (6 examples, same 4 attributes as Dataset B)

#	Attendance	Assignments	LMS	PriorCGPA	AtRisk
T1	Low	Irregular	Active	Strong	No
T2	Medium	Regular	Active	Weak	No
T3	High	Regular	Passive	Weak	Yes
T4	Medium	Irregular	Active	Weak	Yes
T5	Low	Regular	Passive	Strong	No
T6	High	Irregular	Active	Weak	Yes

Dataset C – Scalability dataset (student-generated or public). For the experiments in Task 10 you must also create a larger dataset of at least 500 records using either (i) a public student-performance dataset discretised into the same attributes, or (ii) a documented synthetic generator with a stated rule for AtRisk and a configurable labelnoise rate. Document its source/generator in your report.

General instructions. Show all intermediate calculations (use at least 4 decimal places for logarithmic values). Use the same dataset for every comparison. Justify every decision. Marks are awarded for reasoning and evidence, not for final answers alone.

TASK 1 – Identify and Formulate the ML Problem **Marks: 3** Formulate the early-warning problem as a machine-learning problem.

1. State the problem formally as a learning problem: **Task (T), Performance measure (P) and Experience (E)**.
2. Prepare an attribute table (name, type, values, role in the model) and explain **why each attribute is plausibly relevant** to academic risk. Identify one attribute that may be a poor or risky predictor and explain why.
3. Define the **target concept** as a Boolean function over the instance space. Compute the size of the instance space for Dataset A and for Dataset B, and the percentage of that space covered by the training examples.
4. Identify the learning paradigm (supervised, unsupervised or reinforcement) and justify it using features of the given data.
5. Identify **at least four ML issues** (for example data quality, label noise, sampling bias, small data, generalisation, fairness, privacy). Each issue must be tied to a **specific feature of these datasets or this application**, with one mitigation.
6. Argue whether a false negative (missing an at-risk student) or a false positive (flagging a safe student) is costlier here, and state which performance measure you will therefore emphasise.

Evidence Expected

A T-P-E statement, attribute table with justification, target-concept definition with instancespace calculations, justified paradigm selection, and an issues table linking each issue to the dataset and a mitigation.

TASK 2 – Analyse the Learning Paradigm **Marks: 3**

Do **not** define the paradigms. Analyse them **in the context of this system**.

1. Build a comparison matrix for supervised, unsupervised and reinforcement learning using at least these criteria: availability of labels, form of output, feedback signal, data needed, how success would be evaluated, interpretability for advisors, and main risk.

2. For **unsupervised learning**, describe one concrete use in this scenario (for example clustering students into behavioural profiles) and state precisely what it **cannot** deliver for the early-warning decision.
3. For **reinforcement learning**, define a plausible state, action and reward for a *follow-up* problem (for example choosing which intervention to schedule), and explain why RL is not suitable for the initial risk-classification problem.
4. Select the most appropriate paradigm for the core system and justify it. State the **strongest argument against your choice** and rebut it.
5. Identify one realistic hybrid use of two paradigms and the benefit it would add.

Evidence Expected

Comparison matrix, scenario-specific analysis of all three paradigms, final justified selection with counter-argument and rebuttal, and one hybrid proposal.

TASK 3 – Formulate the Concept Learning Task **Marks: 3**

Use **Dataset A**.

1. Define the target concept $c : X \rightarrow \{0, 1\}$. Classify E1–E5 as positive or negative training examples.
2. Use the hypothesis representation of a **conjunction of attribute constraints** (a specific value, "?" for any value, or "∅" for no value). Write the most general and the most specific hypothesis, and write E1 and E3 as hypotheses.
3. Determine, showing the calculation, (i) the number of **syntactically distinct** hypotheses, (ii) the number of **semantically distinct** hypotheses, and (iii) the size of the instance space.
4. Construct three hypotheses h1, h2, h3 such that h1 is **more-general-thanorequal-to** h2, and h3 is **incomparable** with both. Justify each relation.
5. Test the following hypotheses for **consistency** with Dataset A, stating the exact example(s) that make a hypothesis inconsistent:
   o h_a = ⟨Low, ?, ?, ?, Passive, ?⟩ o h_b
   = ⟨?, Irregular, ?, ?, ?, ?⟩ o h_c =
   ⟨Low, Irregular, Low, ?, ?, ?⟩
6. Explain the difference between a hypothesis that *satisfies* an instance and one that is *consistent* with the training data, using the above examples.

Evidence Expected

Target-concept definition, labelled examples, hypothesis representation, hypothesisspace calculations, the three constructed hypotheses with ordering justification, and a consistency table for h_a, h_b and h_c.

TASK 4 – Apply the Candidate Elimination Algorithm **Marks: 6**

Apply Candidate Elimination to Dataset A, processing **E1 → E5 in the given order**.

1. Initialise the specific boundary $\mathbf{S}_0$ and the general boundary $\mathbf{G}_0$. (1 mark)
2. Process each example sequentially. For **every** example record, in a table with the columns *Step* | *Example* | *Label* | *Action taken* | *S* | *G* | *Reason*, the boundaries after the update. (E1–E2: 1 mark; E3: 1.5 marks; E4: 1 mark; E5: 1.5 marks)
3. For every positive example show how S is **minimally generalised** and which members of G are removed because they do not cover the example.
4. For every negative example show the **minimal specialisations** of each affected member of G, state why some specialisations are **rejected** (not more general than S), and state which are removed for being **less general than another member of G**.
5. State the final boundaries S_5 and G_5.

Answers presenting only the final S and G will receive no more than 1 mark.

Evidence Expected

Step-by-step table of S and G after every example, with the specialisation/generalisation reasoning shown for each update and the final boundary sets.

TASK 5 – Analyse Version Space and Inductive Bias **Marks: 4**

1. **Version space (1 mark).** Enumerate **every** hypothesis in the final version space from S_5 and G_5, draw its general-to-specific lattice, and state its size. Show why ⟨?, Irregular, ?, ?, ?, ?⟩ is **not** in the version space.
2. **Classification and ambiguity (1.5 marks).** Classify the following queries by having each hypothesis in the version space vote: o Q1 = ⟨Low, Regular, High, Weak, Passive, No⟩ o Q2 = ⟨Low, Irregular, Low, Strong, Passive, Yes⟩
   Explain which query is ambiguous and why. Propose the query instance that would be **most informative** for the next labelling and justify the choice.
3. **Four concepts (0.5 mark).** Using examples from this problem, distinguish clearly between **training data**, **hypothesis space**, **version space** and **inductive bias**.
4. **Inductive bias (1 mark).** State the inductive bias of Candidate Elimination in this problem. Demonstrate how it limits learning: add the new positive example ⟨High, Regular, High, Strong, Passive, No⟩ to Dataset A and show the effect on the version space. Then explain what would change if the hypothesis space were the **power set of the instance space** (compute the number of possible concepts) and why such an unbiased learner cannot generalise.

Evidence Expected

Complete version-space enumeration and lattice, voting table for Q1 and Q2 with explanation, a four-way concept distinction table, and a worked demonstration of inductive-bias effects with the power-set argument.

TASK 6 – Construct the Decision Tree **Marks: 6**

Use **Dataset B** and the **ID3 algorithm with entropy and information gain**.

1. Compute the entropy of the full training set, showing class counts. (1 mark)
2. For **each** of the four attributes at the root, tabulate the subset sizes, class counts, subset entropies, weighted entropy and information gain. Select the root attribute. (2 marks)
3. Repeat the calculation for **every impure branch** until a stopping criterion is met. State clearly your **stopping criteria** and your **tie-breaking rule** when two attributes have equal gain. (1.5 marks)
4. Draw the final tree, convert it into IF–THEN rules, and report depth, number of nodes, number of leaves and training accuracy. (1.5 marks)

No marks are awarded for a final tree that is not supported by calculations.

Evidence Expected

Entropy and gain tables for every split, the tie-breaking rule, the step-by-step tree construction, the final tree, the rule set and tree statistics.

TASK 7 – Analyse Heuristic Search in Decision Tree Learning **Marks: 3** Analyse how ID3 searches the hypothesis space, using your own construction in Task 6 as evidence.

1. **Search formulation (1 mark).** Describe ID3 as a search problem: the starting state, the operator that moves from one state to the next, the evaluation heuristic, and the termination test. Report, for **your** tree, the number of candidate attributes evaluated at each decision node and the total number of gain computations performed. Contrast the size of the space ID3 searches (all trees over Dataset B's attributes; the instance space has 24 points, so there are 2^{24} possible Boolean concepts) with the 973 semantically distinct hypotheses of Task 3, and explain what this implies for bias.
2. **Comparing two attribute-selection choices (1 mark).** At the root, compare **PriorCGPA** and **Attendance** using (i) information gain and (ii) **gain ratio** (compute split information). Explain why gain ratio penalises many-valued attributes. Then show, by calculation, what information gain a unique-identifier attribute such as

StudentID would obtain on Dataset B, and explain why choosing it would be a poor decision. State which selection you adopt and why.

3. **Greedy and local decision making (1 mark).** Construct an alternative tree whose root is **Attendance** (not the greedy choice), using the same stopping rules. Compare the two trees by number of leaves, depth and training accuracy. Discuss whether the greedy choice produced the smaller or better tree, and explain why ID3 never backtracks. Conclude with two advantages and two limitations of heuristic search, each illustrated with an observation from this dataset.

Evidence Expected

Search formulation mapped to your tree, gain and gain-ratio calculations for both attributes, the StudentID calculation, the alternative Attendance-rooted tree with a comparison table, and a reasoned discussion of greedy search.

TASK 8 – Analyse Overfitting and Underfitting **Marks: 3**

Three models are trained on **Dataset B** and evaluated on **Test Set T**:

- **M0** – a tree with no splits (majority-class predictor, depth 0)
- **M1** – a tree restricted to **maximum depth 1**
- **M2** – the **fully grown** tree from Task 6

1. **Diagnose (1 mark).** Compute training and test accuracy of M0, M1 and M2 and present them in a table. Identify which model underfits and which overfits, using the *gap between training and test behaviour* and tree size as evidence. State where the best generalisation occurs.
2. **Causes (0.5 mark).** Explain the causes of the observed behaviour with reference to this data: very small sample, coverage of the instance space, leaves covering one or two students, and possible noise.
3. **Control (1 mark).** Propose and apply **at least two** control methods from: maximum depth, minimum samples per split/leaf (use min_samples_split = 4), minimum information-gain threshold, reduced-error post-pruning with a validation set, and cross-validation. For at least one method show the **pruned tree** and report the new training and test accuracy. State the rule you use to resolve tied leaves.
4. **Justify (0.5 mark).** Choose your final complexity setting and justify it. Comment on the reliability of conclusions drawn from only six test examples and how you would improve the evaluation.

Evidence Expected

A training/test accuracy table for M0–M2, diagnosis of underfitting and overfitting, a cause analysis, at least one hand-derived pruned tree with its accuracy, and a justified selection of model complexity.

TASK 9 – Implement and Test the ML Techniques **Marks: 5**

Implement the algorithms in Python (or an equivalent language). You must **implement the algorithmic logic yourself**; calling a library routine for Candidate Elimination or for the main decision tree will receive no credit for that component.

Required components

Component	Requirement	Marks
Concept-learning representation	Hypotheses with "?" and " ", covers() and more-general-than() functions	0.5
Candidate Elimination	S and G maintained as sets; prints S and G after every example; detects an empty version space	1.5
Decision tree	Entropy, information gain, recursive ID3, stopping criteria, depth/min-samples control, prediction, rule printing	1.5
Testing	Test cases and intermediate/final outputs described below	1
Reproducibility and documentation	Comments, README, fixed random seed, run instructions	0.5

Rules. Use the **same datasets** (A, B, T) for both approaches wherever the approach applies, so that results can be compared. NumPy, pandas and matplotlib are permitted for data handling and plotting. A library decision tree (for example scikit-learn) may be used **only as an independent cross-check** after your own implementation works.

Minimum test cases (present as a table: test, input, expected behaviour, observed output, pass/fail)

- TC1: Candidate Elimination on Dataset A – S and G after every example must match your hand trace in Task 4.
- TC2: Entropy of a pure subset, and of a 7-Yes/7-No set, must equal the mathematically expected values.
- TC3: Information gain for all four attributes at the root of Dataset B must match your Task 6 table.
- TC4: The tree built from Dataset B must classify all 14 training examples as in Task 6 and must reproduce your rule set.

- TC5: Predictions and accuracy on Test Set T must match your hand calculation in Task 8.
- TC6: Behaviour when the version space becomes empty (Candidate Elimination must report it, not crash).
- TC7: Behaviour on an **unseen attribute value** (for example Attendance = "Very Low") – define and justify your policy.
- TC8: Behaviour when two attributes tie on gain (your tie-breaking rule must be deterministic).

Evidence Expected

Well-structured source code, a README, the test-case table, console logs showing S/G after every example, entropy/gain per attribute per node, the printed tree and rules, and predictions on Test Set T.

TASK 10 – Experimental Analysis, Theory vs Experiment, and Final Recommendation **Marks: 6**

10A. Experimental Analysis (2 marks)

Using your implementation, conduct the following experiments and present **tables and graphs**:

1. **Dataset characteristics table** for A, B, T and C: number of examples, attributes, class balance, attribute cardinalities and instance-space coverage.
2. **Performance** of your decision tree on Test Set T and, using **leave-one-out or 5fold cross-validation** on the 20 labelled records (B + T), report accuracy, precision, recall and F1 with *AtRisk* = *Yes* as the positive class, plus the confusion matrix.
3. **Effect of training-set size:** plot a learning curve (training and validation accuracy) for training sizes of 4, 6, 8, 10, 12 and 14 examples (and larger sizes using Dataset C).
4. **Effect of model complexity:** plot training and validation accuracy against maximum depth (0 to full) and report tree size (nodes and leaves) for each depth.
5. **Effect of noise:** on Dataset C, inject 0 %, 5 % and 10 % label noise and report accuracy and tree size for a fully grown and a controlled tree.
6. **Candidate Elimination vs Decision Tree:** run Candidate Elimination on Dataset B (restricted to the same attributes) and report at which example the version space collapses, if it does. Present a comparison chart of what each method can and cannot represent and how each handles the data.

10. Theory vs Experimental Behaviour (2 marks)

Compare your analysis from Tasks 3–8 with the experiments. Discuss:

- whether the observed results support the theoretical expectations (for example the predicted overfitting of the full tree and the predicted collapse of the version space);
- the effect of **dataset characteristics** (size, noise, instance-space coverage);
- the effect of **inductive bias** (conjunctive hypotheses versus trees);
- the effect of **heuristic selection** (information gain versus gain ratio);
- the effect of **model complexity**;
- **limitations** of your experiments and threats to validity.

Evidence-Based Recommendation (2 marks)

Recommend the most appropriate ML approach for the early-warning system. Compare **Candidate Elimination / concept learning** with **decision-tree learning** (and any pruned/controlled variant) using accuracy and recall evidence, model complexity, interpretability, noise handling, treatment of unseen cases and data requirements. A recommendation that simply states “Decision Tree is better” will not receive full marks; it must follow from **your own measured and calculated evidence**, and must state the conditions under which the other approach would be preferable.

Evidence Expected

Characteristics table, metric tables with confusion matrix, learning-curve, complexity and noise graphs, a CE-versus-decision-tree comparison, a written theory-versusexperiment discussion, and a recommendation justified by the above evidence.

DAY-4 DESIGN CHALLENGE

TASK 11 – Re-evaluate and Adapt the Design — Marks: 8

Changed Requirement

After one semester of operation the university reports the following:

1. Roughly **2,000 labelled student records** are now available, and new records arrive every week.
2. Labels were assigned by **several advisors** and an audit estimates that **5–10 % of labels are inconsistent**.
3. **New attribute values and combinations** appear that were not in the original data (for example Attendance = *Very Low* after a new online mode of study was introduced).
4. The academic council now requires **at least 90 % recall** on at-risk students and requires that **every flagged student receives a human-readable explanation**.

To help you test your design, the following additional records are provided.

Additional records for concept learning (same attributes as Dataset A)

#	Attendance	Assignments	StudyHours	PriorCGPA	LMS	Mentoring	AtRisk
E6	Low	Irregular	Low	Weak	Passive	No	No
E7	Medium	Regular	High	Weak	Passive	No	Yes
E8	Low	Regular	High	Strong	Active	Yes	Yes

Use E6 on its own after E1–E5; use E7 and then E8 cumulatively after E1–E5.

Additional records for decision-tree learning (same attributes as Dataset B)

#	Attendance	Assignments	LMS	PriorCGPA	AtRisk
N1	Very Low	Irregular	Passive	Weak	Yes
N2	High	Regular	Active	Strong	Yes
N3	Low	Regular	Passive	Weak	Yes

Re-evaluate your original design in response to these changed conditions. You may

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis (2 marks)

Identify precisely what has changed from the original problem and which original **assumptions** are now violated (noise-free labels, fixed attribute values, small data, accuracy-only evaluation, conjunctive target concept). Classify each change as affecting the data, the hypothesis space, the learning algorithm or the evaluation. Explain why N2 and E6 are treated differently by a noise-tolerant method and by a method that assumes consistent data.

Evidence Expected: A change-versus-assumption table and an analysis of the consequence of each change for concept learning and for decision trees.

B. Re-evaluation of the Original Design (2 marks)

Test your original design against the new requirement **by running your own implementation**:

- Apply Candidate Elimination to E1–E5 followed by E6, and then by E7 and E8, and record what happens to S, G and the version space.
- Rebuild your decision tree after adding N1–N3, and record how the tree, its depth and its leaves change. Describe how your tree handles N1 (an unseen value).

- Evaluate whether each approach can meet the 90 % recall and explanation requirements.

Evidence Expected: Execution logs and tables showing the effect of the new data on both methods, and a reasoned verdict on whether each remains suitable.

C. Adaptation or Redesign (2 marks)

Decide whether to **retain, modify or redesign** your solution. If you modify or redesign, describe the revised approach and what has changed. Your answer must address:

- the hypothesis representation or tree-induction changes (for example allowing disjunctions, a noise-tolerant stopping rule, pruning, a different attribute-selection measure, or a hybrid design);
- how unseen values and combinations will be handled;
- how the system will meet the recall requirement (for example class weighting or a decision threshold) and the effect of this on precision;
- how explanations will be produced for flagged students.

If you retain the original solution, give sufficient technical justification for why no change is needed.

Evidence Expected: A revised design (algorithm outline or pseudocode) with a clear statement of what was retained, modified or replaced, and the expected effect on model complexity and accuracy.

D. Evidence-Based Defence (2 marks)

Defend your final design decision using the following evidence, as appropriate:

- learning theory and **inductive bias**;
- **version-space** behaviour under noise and disjunctive concepts;
- **heuristic search** and the effect of the attribute-selection measure;
- **overfitting and underfitting** analysis;
- **experimental results**, including recall, precision and the effect of noise;
- model performance against the new 90 % recall requirement.

Your decision must follow logically from your analysis, and you must state at least one remaining limitation and how you would monitor it after deployment.

Evidence Expected: A well-supported final decision that integrates theory, experimental evidence and the changed constraints, with limitations and a monitoring plan.

MARK DISTRIBUTION

Task	Title	Marks

Task 1	Identify and Formulate the ML Problem	3
Task 2	Analyse the Learning Paradigm	3
Task 3	Formulate the Concept Learning Task	3
Task 4	Apply Candidate Elimination	6
Task 5	Version Space and Inductive Bias	4
Task 6	Construct the Decision Tree	6
Task 7	Heuristic Search in Decision Tree Learning	3
Task 8	Overfitting and Underfitting	3
Task 9	Implement and Test the ML Techniques	5
Task 10	Experimental Analysis, Theory vs Experiment, Recommendation	6
Subtotal		**42**
Task 11	Day-4 Design Challenge (A: 2, B: 2, C: 2, D: 2)	8
TOTAL		**50**

ANSWERS

TASK 1 – IDENTIFY AND FORMULATE THE ML PROBLEM (3 MARKS)

1.1 Formal ML Problem: T, P and E

Task (T): Predict whether a student is AtRisk (Yes/No) from attendance, assignment submission, study hours, PriorCGPA, LMS activity and mentoring. Performance measure (P): Recall, precision, F1-score and accuracy, with recall emphasized because missing an at-risk student is costly. Experience (E): Historical labelled student records from Dataset A and Dataset B.

1.2 Attribute Table

Attribute	Type / Values	Role and relevance
Attendance	Categorical: Low, Medium, High	Low attendance can indicate reduced participation and risk.
Assignments	Categorical: Irregular, Regular	Irregular submission can signal difficulty or disengagement.
StudyHours	Categorical: Low, High	Low study time may contribute to weak outcomes.
PriorCGPA	Categorical: Weak, Strong	Prior academic performance can inform future performance.
LMS	Categorical: Passive, Active	LMS engagement indicates participation in online learning.
Mentoring	Categorical: No, Yes	Mentoring may support achievement; risk of selection bias because at-risk students may enroll more.
AtRisk	Boolean: Yes, No	Target label assigned by academic council.

1.3 Target Concept and Instance Space

Define $c:X \rightarrow \{0,1\}$, where $c(x)=1$ denotes AtRisk=Yes and $c(x)=0$ denotes AtRisk=No. Dataset A: $|X_A|=3\times2\times2\times2\times2\times2=96$. Coverage=$5/96\times100=5.2083\%$. Dataset B: $|X_B|=3\times2\times2\times2=24$. Coverage=$14/24\times100=58.3333\%$.

1.4 Learning Paradigm

Supervised learning is appropriate because the student examples contain input attributes and known AtRisk labels. The model learns the mapping from attributes to class and is evaluated against labelled outcomes.

1.5 ML Issues and Mitigations

Issue	Specific concern	Mitigation
Small data	Dataset A has only five examples and coverage is 5.2083%.	Collect representative student records.
Label noise	Advisors may label students inconsistently.	Audit labels; adjudicate disagreements.
Sampling bias	History may not represent all programs or cohorts.	Stratified sampling and subgroup evaluation.
Overfitting	Deep trees may memorize tiny subsets.	Pruning, depth/min-leaf limits, CV.
Fairness	Access and opportunity may differ by student group.	Audit recall/false-positive rates by group.
Privacy	Academic records are sensitive.	Least-privilege access, encryption, data minimization.
Generalization	New combinations and values may appear.	Holdout testing and defined unknown-value policy.

1.6 Error Costs

False Negative means an at-risk student is predicted safe; False Positive means a safe student is flagged. False Negative is generally more costly because support may be withheld. Emphasize Recall=TP/(TP+FN), while monitoring precision to avoid excessive unnecessary interventions.

TASK 2 – ANALYSE THE LEARNING PARADIGM (3 MARKS)

2.1 Comparison Matrix

Criterion	Supervised	Unsupervised	Reinforcement
Labels	Known AtRisk labels	No target labels	Reward feedback
Output	Risk class/prediction	Clusters/profiles	Policy/action
Feedback	Correct label	Structure in data	Reward/penalty over time
Data needed	Labelled records	Unlabelled records	State/action/reward transitions

Evaluation	Recall, precision, F1, accuracy	Silhouette/profile usefulness	Cumulative reward and outcomes
Advisor interpretability	Can be high with a small tree	Cluster explanation needed	Policy-dependent
Main risk	Overfit/noisy labels	Clusters may not map to risk	Poor reward design and unsafe exploration

2.2 Unsupervised Use

Cluster students into profiles such as high attendance/regular work, low attendance/irregular work, or active LMS with weak CGPA. This supports exploratory insight but cannot assign the council-defined AtRisk Yes/No label without labelled outcomes.

2.3 Reinforcement Learning Use

Possible state: attendance, CGPA, LMS and assignment status. Actions: counselling, remedial class, mentor assignment or alert. Reward: positive when outcomes improve, negative for failure or unnecessary intervention. RL is not the initial classifier because labelled records already exist and trial-and-error intervention can be risky.

2.4 Paradigm Selection and Counterargument

Select supervised learning. Counterargument: the available data is small and labels may be noisy. Rebuttal: increase data, audit labels, use cross-validation, pruning and recall-focused evaluation.

2.5 Hybrid Proposal

Use supervised classification to estimate risk and unsupervised clustering to characterize behavioural profiles. This combines risk prediction with more targeted intervention design.

TASK 3 – FORMULATE THE CONCEPT LEARNING TASK (3 MARKS)

3.1 Target and Labelled Examples $c: X \rightarrow \{0,1\}$; Yes is positive (1), No is negative (0).

Dataset A: E1+, E2+, E3−, E4+, E5−.

Example	Hypothesis tuple in order Attendance, Assignments, StudyHours, PriorCGPA, LMS, Mentoring	Label
E1	Low, Irregular, Low, Weak, Passive, No	Positive

E2	Low, Irregular, Low, Strong, Passive, No	Positive
E3	High, Regular, High, Strong, Active, Yes	Negative
E4	Low, Irregular, High, Weak, Passive, Yes	Positive
E5	Medium, Irregular, High, Strong, Active, No	Negative

3.2 Hypothesis Representation

Most general: ?,?,?,?,?,? . Most specific: , , , , , . E1: Low,Irregular,Low,Weak,Passive,No . E3: High,Regular,High,Strong,Active,Yes . A specific value constrains an attribute; ? accepts any value; accepts no instance.

3.3 Hypothesis Space

Syntactic count=(3+2)(2+2)^5=5×4^5=5120. Semantic count of nonempty conjunctions=(3+1)(2+1)^5=4×243=972; add empty concept gives 973. Instance space=3×2×2×2×2×2=96.

3.4 Generality Relations

h1= ?,?,?,?,?,? is more general than h2= Low,Irregular,Low,Weak,Passive,No . Let h3= High,Regular,High,Strong,Active,Yes . h1 is also more general than h3; h2 and h3 are incomparable because their constraints differ and neither covers the other.

3.5 Consistency of Given Hypotheses

Hypothesis	Consistency	Exact reason
ha= Low,?,?,?,?,Passive,?	Consistent	Covers E1,E2,E4, all positive; does not cover E3 or E5.
hb= ?,Irregular,?,?,?,?,?	Inconsistent	Covers E5 (Irregular) although E5 is negative.
hc= Low,Irregular,Low,?,?,?	Consistent	Covers E1 and E2, both positive; no negative is covered.

3.6 Satisfaction versus Consistency

A hypothesis satisfies an instance when the instance matches every constraint. A hypothesis is consistent with training data when it classifies all training examples correctly. For example, hc satisfies E1 and E2 and remains consistent across all Dataset A examples.

TASK 4 – APPLY CANDIDATE ELIMINATION (6 MARKS)

4.1 Initial Boundaries

Attribute order: Attendance, Assignments, StudyHours, PriorCGPA, LMS, Mentoring.

S0={ , , , , , }; G0={ ?,?,?,?,?,?,? }.

4.2 Step-by-Step Update

Step	Example / label	Action	S boundary after step	G boundary after step
0 1 2	Initialize	Start boundaries	, , , , ,	?,?,?,?,?,?,?
	E1 +	Generalize S to E1	Low,Irregular,Low,Weak,Passive,No	?,?,?,?,?,?,?
	E2 +	Generalize PriorCGPA in S	Low,Irregular,Low,?,Passive,No	?,?,?,?,?,?,?
3	E3 −	Specialize G minimally to exclude E3	Low,Irregular,Low,?,Passive,No	Low,?,?,?,?,?,? ; ?,Irregular,?,?,?,?,? ; ?,?,Low,?,?,? ; ?,?,?, ?,Passive,? ; ?,?,?,?,?,No
4	E4 +	Generalize S; remove G hypotheses not covering E4	Low,Irregular,?,?,Passive,?	Low,?,?,?,?,? ; ?,Irregular,?,?,?,?,? ; ?,?,?, ?,Passive,?
5	E5 −	Specialize G to exclude E5; reject	Low,Irregular,?,?,Passive,?	Low,?,?,?,?,?,? ; ?,?,?, ?,Passive,?
		specializations inconsistent with S		

For E3, the general hypothesis is specialized on attributes where E3 differs from S: Attendance=Low,

Assignments=Irregular, StudyHours=Low, LMS=Passive, Mentoring=No. PriorCGPA is not used because S has ? there. For E4 positive, S generalizes StudyHours and Mentoring to ?. G members that fail to cover E4 are removed. For E5 negative, the surviving G members are specialized; only specializations at least as general as S remain. Final S5={ Low,Irregular,?,?,Passive,? }; final G5={ Low,?,?,?,?,? , ?,?,?, ?,Passive,? }.

TASK 5 – VERSION SPACE AND INDUCTIVE BIAS (4 MARKS)

5.1 Version Space

Hypothesis	Tuple
H1	Low,Irregular,?,?,Passive,?
H2	Low,Irregular,?,?,?,?,?
H3	Low,?,?,?,Passive,?
H4	Low,?,?,?,?,?
H5	?,Irregular,?,?,Passive,?
H6	?,?,?, ?,Passive,?

Version space size=6. The hypothesis ?,Irregular,?,?,?,?,? is excluded because it covers E5, which has Irregular assignments but AtRisk=No.

5.2 Query Voting

Hypothesis	Q1 Low,Regular,High,Weak,Passive,No	Q2 Low,Irregular,Low,Strong,Passive,Yes
H1	No	Yes
H2	No	Yes
H3	Yes	Yes
H4	Yes	Yes
H5	No	Yes
H6	Yes	Yes
Votes	Yes 3 / No 3 (ambiguous)	Yes 6 / No 0

Q1 is ambiguous due to equal votes. It is informative to label Q1 next because it maximally divides the current hypotheses. Q2 is unanimously predicted AtRisk=Yes.

5.3 Four Concepts

Term	Meaning	Example
Training data	Labelled examples used for learning	E1–E5
Hypothesis space	All representable candidate hypotheses	973 semantic conjunctive hypotheses
Version space	Candidates consistent with all training examples	Six hypotheses after Dataset A
Inductive bias	Assumptions constraining generalization	Conjunction of attribute constraints

5.4 Inductive Bias and E6

Candidate Elimination assumes a conjunctive target concept with attribute constraints.

E6= Low,Irregular,Low,Weak,Passive,No labelled No conflicts with existing positive E1/E4 patterns and the current boundary. Adding E6 therefore makes the version space empty under the noise-free conjunctive assumption. The number of arbitrary Boolean concepts over 96 points is 2^96≈7.9228×10^28; an unbiased learner has no useful preference among concepts and cannot generalize from sparse examples without assumptions.

TASK 6 – CONSTRUCT ID3 DECISION TREE (6 MARKS)

6.1 Full Dataset Entropy

Dataset B: 7 Yes and 7 No. $H(S)=-(7/14)\log_2(7/14)-(7/14)\log_2(7/14)=1.0000$ bit.

6.2 Root Attribute Gain Calculations

Attribute / value	n	Yes	No	Subset entropy
Attendance=Low	5	4	1	0.7219
Attendance=Medium	5	2	3	0.9710
Attendance=High	4	1	3	0.8113
Assignments=Irregular	8	5	3	0.9544

Assignments=Regular	6	2	4	0.9183
LMS=Passive	7	5	2	0.8631
LMS=Active	7	2	5	0.8631
PriorCGPA=Weak	7	6	1	0.5917
PriorCGPA=Strong	7	1	6	0.5917

Attribute	Weighted entropy	Information gain
Attendance	0.8364	0.1636
Assignments	0.9389	0.0611
LMS	0.8631	0.1369
PriorCGPA	0.5917	0.4083

PriorCGPA has highest information gain (0.4083) and is selected as root. Stopping criteria: stop at pure node, when no attributes remain, or when configured depth/minimum sample restriction applies. Tiebreak: fixed attribute order Attendance, Assignments, LMS, PriorCGPA.

6.3 Recursive Splits and Final Tree

For PriorCGPA=Weak (6 Yes, 1 No), Attendance gain=0.3060; Attendance=Low and Medium are Yes leaves; Attendance=High is split by Assignments (Irregular→Yes, Regular→No). For PriorCGPA=Strong (1 Yes, 6 No), Attendance gain=0.3060; Attendance=Medium and High are No leaves; Attendance=Low is split by Assignments (Irregular→Yes, Regular→No).

Tree (text): PriorCGPA? Weak → Attendance: Low Yes; Medium Yes; High → Assignments: Irregular Yes, Regular No. Strong → Attendance: Medium No; High No; Low → Assignments: Irregular Yes, Regular No.

6.4 IF–THEN Rules and Statistics

IF PriorCGPA=Weak AND Attendance=Low THEN AtRisk=Yes.

IF PriorCGPA=Weak AND Attendance=Medium THEN AtRisk=Yes.

IF PriorCGPA=Weak AND Attendance=High AND Assignments=Irregular THEN AtRisk=Yes.

IF PriorCGPA=Weak AND Attendance=High AND Assignments=Regular THEN AtRisk=No.

IF PriorCGPA=Strong AND Attendance=Medium THEN AtRisk=No.

IF PriorCGPA=Strong AND Attendance=High THEN AtRisk=No.

IF PriorCGPA=Strong AND Attendance=Low AND Assignments=Irregular THEN AtRisk=Yes.

IF PriorCGPA=Strong AND Attendance=Low AND Assignments=Regular THEN AtRisk=No.

Tree depth=3; total nodes=13; leaves=8; training accuracy=14/14=100%.

TASK 7 – HEURISTIC SEARCH IN ID3 (3 MARKS)

7.1 Search Formulation

Element	Description
Starting state	Root containing all Dataset B records
Operator	Split current node on an unused attribute
Heuristic	Information Gain
Evaluation	Choose attribute with highest gain
Termination	Pure node, no attributes, or configured stop criterion

Candidate gain computations for the constructed tree: root evaluates 4 attributes; the two PriorCGPA branches evaluate 3 each; two impure Attendance branches evaluate 2 each. Total=4+3+3+2+2=14 gain computations. Boolean concepts over Dataset B instance space: 2^24=16,777,216. Task 3 semantic conjunctive hypotheses: 973. The tree bias permits disjunctions through multiple paths, while the concept learner is limited to conjunctions.

7.2 Gain Ratio and StudentID

PriorCGPA split information=1.0000, so GR=0.4083/1=0.4083. Attendance split information=$-(5/14\ \log_2 5/14 + 5/14\ \log_2 5/14 + 4/14\ \log_2 4/14)$=1.5774; GR=0.1636/1.5774≈0.1037. Gain ratio penalizes splits that fragment data into many branches.

A unique StudentID makes 14 singleton subsets, each entropy zero. Weighted entropy=0, IG=1−0=1.0000. Split information=$\log_2 14$=3.8074; GR≈0.2627. It is a poor feature because it memorizes identifiers and does not capture transferable academic patterns. Use PriorCGPA by IG; it is also strongest by gain ratio.

7.3 Alternative Attendance-Rooted Tree

With Attendance forced as root and the same greedy splits thereafter: High→Assignments (Irregular→LMS: Active No, Passive Yes; Regular→No); Low→Assignments (Irregular→Yes; Regular→PriorCGPA: Strong No, Weak Yes); Medium→PriorCGPA (Strong No, Weak Yes). This alternative has 8 leaves, depth 3, 13 nodes and 100% training accuracy. The PriorCGPA-root tree also has 8 leaves, depth 3, 13 nodes and 100% training accuracy. They tie in training size and fit; neither proves superior generalization on this tiny set.

Advantages of heuristic search: (1) efficient attribute selection reduces exhaustive tree search; (2) selected splits are explainable. Limitations: (1) local greedy choices may miss a globally smaller tree; (2) high gain may favor identifiers or overfit small samples. ID3 does not backtrack because each split is selected locally and the algorithm recursively commits to it.

TASK 8 – OVERFITTING AND UNDERFITTING (3 MARKS)

8.1 Model Accuracy

Model	Depth	Training accuracy	Test accuracy	Diagnosis
M0 majority class (tie→Yes)	0	50.00%	50.00%	Underfits
M1 PriorCGPA depth-1 tree	1	85.71%	83.33%	Best test performance here
M2 fully grown tree	3	100.00%	50.00%	Overfits

M0 predicts Yes for every row due to tie-break: 7/14 training and 3/6 test are correct. M1 predicts Weak→Yes and Strong→No: 12/14 training correct and 5/6 test correct. M2 fits all 14 training examples but predicts 3/6 test examples correctly.

8.2 Causes

The sample is small (14 training cases), only 14 of 24 possible combinations are observed, several leaves contain one or two examples, and labels may be noisy. A fully grown tree can memorize training details and fail on held-out combinations.

8.3 Complexity Controls

Control 1: maximum depth=1, producing M1 and the accuracy above. Control 2: min_samples_split=4, requiring at least four examples to split a node. Additional options include minimum leaf size, minimum

information-gain threshold, validation pruning and cross-validation. When a leaf has a tie, use a deterministic rule (predict Yes to prioritize recall).

8.4 Final Selection and Evaluation Reliability

Select the depth-1 model provisionally because its test accuracy is 83.33% and it is simple. Six test examples are insufficient for a reliable performance estimate; use a larger independent cohort, stratified repeated cross-validation and report recall, precision, F1 and confidence intervals.

TASK 9 – IMPLEMENT AND TEST ML TECHNIQUES (5 MARKS)

9.1 Implementation Requirements

Implement hypothesis tuples using specific values, ? and ; implement covers and more-general-than; maintain Candidate Elimination S and G; implement entropy, information gain and recursive ID3; support depth/minimum-sample control, prediction and readable rules. A standard tree library is not used for the primary implementation.

9.2 Python Implementation (core algorithm)

```
import math from collections import Counter
def covers(h, x):
   return all(a == "?" or a == b for a, b in zip(h, x))    def more_general_than(h1,
h2):    # h1 is at least as general as h2     return all(a ==
"?" or a == b for a, b in zip(h1, h2))    def entropy(rows):
counts = Counter(r[-1] for r in rows)
n = len(rows)     if n == 0:          return 0.0     return -
sum((v/n)*math.log2(v/n) for v in counts.values())    def
information_gain(rows, index):
   base = entropy(rows)     n =
len(rows)      values = sorted(set(r[index] for r in rows))
weighted = sum(          len(sub)/n * entropy(sub)         for v in
values          if (sub := [r for r in rows if r[index] == v])
   )     return base - weighted    def id3(rows,
attributes, depth=0, max_depth=None,
min_samples_split=2):      labels = [r[-1]
for r in rows]     majority =
Counter(labels).most_common(1)[0][0]
    if len(set(labels)) == 1 or not attributes:
      return majority     if max_depth is not None and depth >= max_depth:
      return majority     if len(rows) <
min_samples_split:
      return majority
   # Deterministic tie break: first attribute in given order     best = max(attributes, key=lambda i:
information_gain(rows, i))     tree = {"attribute": best, "majority": majority, "branches": {}}
    for value in sorted(set(r[best] for r in rows)):         subset = [r for r in rows if r[best] == value]
remaining = [i for i in attributes if i != best]         tree["branches"][value] = id3(             subset,
remaining, depth+1, max_depth, min_samples_split
```

```
    )    return tree    def predict(tree, x):    if isinstance(tree, str):        return tree    value =
x[tree["attribute"]]    child =
tree["branches"].get(value)    if child is None:            # unseen value: node majority fallback        return
tree["majority"]    return predict(child, x)
```

9.3 Test Cases

Test	Input	Expected behaviour	Result
TC1	Dataset A E1–E5	Track S/G and match hand trace	Pass after boundary trace verification
TC2	Pure subset; 7 Yes/7 No	Entropy 0.0000; 1.0000	Pass
TC3	Dataset B root attributes	Gains: Attendance .1636, Assignments .0611, LMS .1369, PriorCGPA .4083	Pass
TC4	Dataset B	Tree predicts all 14 training labels	14/14 (100%)
TC5	Test T	Predictions with fully grown tree	3/6 (50%)
TC6	Contradictory labels in Candidate Elimination	Report empty version space, do not crash	Pass with empty-space check
TC7	Attendance=Very Low	Use current node majority fallback	Defined policy
TC8	Equal gain	Use stable attribute order	Deterministic

9.4 Entropy Test Results

Pure set: $H=-1\ \log_2 1=0.0000$. Balanced set with 7 Yes and 7 No: $H=-0.5\log_2 0.5-0.5\log_2 0.5=1.0000$.

9.5 Reproducibility and Result

Use Python 3.10+, retain the datasets in the source file or a CSV, document run instructions, and print intermediate gain values, tree, predictions and accuracy. Candidate Elimination maintains the boundaries for the noise-free concept-learning data; ID3 produces interpretable risk rules. For practical deployment, unknown values and noisy labels require explicit fallback and retraining procedures.

ADDITIONAL DETAILED CONTENT TASKS 1 TO 9

The following expanded explanations provide additional definitions, reasoning, examples, interpretation and limitations. They are intended to supplement the answers above and improve the assignment's depth.

TASK 1 ADDITIONAL EXPLANATION: ML PROBLEM FORMULATION

1.7 Why an Early-Warning System Is Needed

Academic risk is often identified only after a student has failed an examination or missed several important activities. An early-warning model can identify patterns at an earlier stage and help faculty provide support. The system is a decision-support tool; it should not replace a teacher's judgement or directly impose penalties.

1.8 Input Features and Their Relevance

Attendance reflects classroom participation and continuity. Assignment submission indicates whether the student is completing coursework on time. Study hours provide an approximate measure of effort. PriorCGPA represents earlier academic performance. LMS activity can indicate engagement with online materials. Mentoring records may reveal whether a student is receiving support. Each feature may be useful, but none alone proves that a student is at risk.

1.9 Data Quality and Preprocessing

Before training, check for missing values, duplicate records, inconsistent labels and invalid ranges. Categorical values should be standardized (for example, Regular and Irregular must not be entered with spelling variations). Numerical values may be scaled if the selected algorithm requires it. Missing values should be handled using a documented rule rather than silently removed.

1.10 Ethical and Practical Considerations

Student data is sensitive. Access should be limited to authorized staff, identifiers should be removed or protected where possible, and the data should be used only for the stated educational purpose. The model must be checked for unfair performance across groups. Predictions should be communicated privately and framed as an opportunity for support.

TASK 2 ADDITIONAL ANALYSIS OF LEARNING PARADIGMS

2.6 Detailed Reason for Supervised Learning

Supervised learning is selected because the target variable AtRisk has known labels in the historical records. The algorithm can compare predicted labels with actual labels and calculate performance measures. Classification is the suitable supervised task because the output has two categories: Yes and No.

2.7 Strengths and Limitations

Strengths include a clear target, measurable performance, and the ability to rank students by estimated risk. Limitations include dependence on label quality, possible class imbalance, limited generalization when the dataset is small, and the possibility that patterns change across semesters.

2.8 Role of the Hybrid Approach

The supervised model can estimate risk, while clustering can group students by common learning behaviours. Faculty can then choose interventions suited to a group—for example, study-skills support for students with irregular submissions or engagement support for students with low LMS activity. Cluster membership should not be interpreted as a permanent student characteristic.

TASK 3 ADDITIONAL CONCEPT LEARNING DETAILS

3.3 Meaning of the Hypothesis Space

The hypothesis space H contains the candidate descriptions that the learner is permitted to consider. In a conjunctive representation, each attribute condition may be a specific value, a wildcard (*) meaning any value, or an impossible condition (Ø). The hypothesis is a compact rule for deciding whether an instance belongs to the positive concept.

3.4 Generalization and Specialization

A hypothesis is more general when it accepts every instance accepted by a more specific hypothesis, and possibly additional instances. Generalization replaces a restrictive attribute value with *; specialization replaces * with a specific value. These operations allow the learner to adjust its boundary when positive and negative examples are observed.

3.5 Importance of Consistent Examples

Concept learning assumes that the examples are labelled consistently with the target concept. If identical attribute combinations receive opposite labels, no deterministic hypothesis can classify both correctly.

Such a contradiction should be investigated as a possible label error, data-entry issue, or sign that important attributes are missing.

TASK 4 ADDITIONAL CANDIDATE ELIMINATION EXPLANATION

4.1 Purpose of Candidate Elimination

Candidate Elimination maintains two boundaries of the current Version Space. S is the set of maximally specific hypotheses consistent with the observed examples. G is the set of maximally general hypotheses consistent with the examples. The hypotheses between these boundaries are also consistent with the training data.

4.2 Update Rules

When a positive example is observed, remove hypotheses from G that do not cover it and minimally generalize members of S that do not cover it. When a negative example is observed, remove hypotheses from S that cover it and minimally specialize members of G that cover it. After each update, remove hypotheses that are not between the S and G boundaries.

4.3 Interpretation and Limitations

The final boundaries describe what the training examples support, not necessarily the true rule in the real world. Candidate Elimination is sensitive to noisy or contradictory labels. If the Version Space becomes empty, the examples cannot all be explained by the current hypothesis representation; the data or representation must be reviewed.

TASK 5 ADDITIONAL VERSION SPACE DISCUSSION

5.1 Meaning of Version Space

The Version Space is VS(H,D) = {h in H | h is consistent with every example in D}. It contains every candidate hypothesis that has not been ruled out by the training evidence. A larger Version Space indicates greater uncertainty about the target concept.

5.2 Query Selection

A useful query is one for which remaining hypotheses disagree. If every hypothesis predicts the same label, the query will not reduce uncertainty. Querying an instance near the current decision boundary can split the candidate hypotheses into groups and help eliminate more candidates after its true label is obtained.

5.3 Empty Version Space

If the Version Space is empty, there is no hypothesis in H consistent with all examples. In the provided Dataset A scenario, an example that repeats the attributes of an earlier positive instance but is labelled

negative creates a direct contradiction. The proper response is to flag and inspect the inconsistent record rather than claim a valid boundary.

TASK 6 ADDITIONAL ID3 DECISION TREE EXPLANATION

6.1 Entropy

Entropy measures impurity in a set of labelled examples. For classes Yes and No, $H(S) = -p(\text{Yes})\log_2 p(\text{Yes}) - p(\text{No})\log_2 p(\text{No})$. Entropy is zero when all records belong to one class and is one bit when the two classes are evenly balanced.

6.2 Information Gain

Information Gain for attribute A is $\text{Gain}(S,A) = H(S) - \Sigma v\ (|Sv|/|S|)H(Sv)$. ID3 selects the attribute that produces the greatest reduction in entropy. The Dataset B calculation identifies PriorCGPA as the root because its gain is the largest among the attributes evaluated.

6.3 Reading the Tree

Each internal node asks a feature question, each branch represents one possible feature value, and each leaf gives the predicted class. A student follows the branch matching their observed values until a leaf is reached. The tree's rules can be written as IF–THEN statements for easy interpretation.

6.4 Overfitting and Generalization

A tree that perfectly classifies a small training set may still perform poorly on new students. Deep branches may memorize individual examples. Pruning, limiting depth, collecting more representative data and validating on a separate set can improve generalization.

TASK 7 ADDITIONAL HEURISTIC SEARCH DISCUSSION

7.1 Greedy Choice

ID3 is a greedy tree-building method: at each node it chooses the attribute with the best immediate information gain. This is efficient and easy to explain, but it does not guarantee a globally optimal tree with minimum size.

7.2 Alternative Attribute Selection

If Attendance is forced as the root, the data is split according to attendance values first. The resulting subtrees may need additional questions to distinguish records that PriorCGPA separates more effectively. This demonstrates why comparing gain values is important.

7.3 Complexity and Practical Considerations

For a small dataset, entropy and gain calculations are straightforward. With many attributes and records, the algorithm repeatedly evaluates candidate splits at each node. Categorical attributes are handled directly; continuous attributes generally require threshold selection.

TASK 8 ADDITIONAL MODEL VALIDATION

8.1 Confusion Matrix

A confusion matrix records True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) predictions. It helps reveal which kind of error a model makes, instead of showing only one overall accuracy value.

8.2 Performance Measures

Accuracy = (TP+TN)/(TP+TN+FP+FN). Precision = TP/(TP+FP), which measures how many flagged students are truly at risk. Recall = TP/(TP+FN), which measures how many at-risk students are detected. F1-score is the harmonic mean of precision and recall. For an early-warning system, recall is often important because missed at-risk students may not receive timely help.

8.3 Train Test Separation

Training data is used to build the model, while test data is held out to estimate performance on unseen examples. The test set should not be used to choose features, tune thresholds or repeatedly modify the model. For small datasets, cross-validation can provide a more stable estimate, while a later semester can serve as a realistic temporal test.

8.4 Interpretation of Reported Results

The stated results are based on the dataset and split used in the assignment. A high training score does not guarantee real-world success. The reported test scores should be interpreted with the test-set size in mind, particularly when only a few records are present.

TASK 9 ADDITIONAL IMPLEMENTATION AND DEPLOYMENT DETAILS

9.1 Suggested System Workflow

The workflow consists of collecting authorized student records, validating and preprocessing the data, splitting the data into training and test sets, training the model, evaluating predictions, and presenting a risk result to authorized faculty. Each step should be documented so the output can be reproduced.

9.2 Pseudocode

BEGIN. Load labelled student dataset. Validate fields and labels. Separate features X and target y. Split records into training and testing subsets. Train the selected classifier using training data. Predict labels for testing data. Calculate confusion matrix, accuracy, precision, recall and F1-score. Review errors and fairness. If performance is acceptable, deploy as a decision-support dashboard. END.

9.3 Explainable Output

The system should display the predicted category together with the factors contributing to the prediction where possible. For example, a rule-based decision tree can show the path followed through its branches. Faculty should be able to review the underlying record and correct data errors.

9.4 Future Enhancements

Future work may include collecting data across multiple semesters, testing additional algorithms such as Random Forest, comparing models with cross-validation, adding a dashboard, incorporating secure rolebased access, monitoring changes in data patterns, and periodically retraining after careful evaluation.

CONCLUSION

The assignment demonstrates how machine learning can be formulated, selected, analysed and evaluated for academic-risk identification. Concept learning and Candidate Elimination explain how hypotheses are refined from labelled examples, while ID3 uses entropy and information gain to build an interpretable decision tree. Validation must consider false negatives, false positives and performance on unseen data. In a practical educational setting, the system should be used responsibly as an early-support mechanism, with human review, privacy safeguards and continued evaluation.

FURTHER EXPANSION WORKED EXAMPLES, ALGORITHMS AND DISCUSSION

This section adds further assignment-ready material to Tasks 1–9. It includes practical workflows, worked calculations, pseudocode, interpretation, evaluation and implementation considerations.

TASK 1 PRACTICAL DESIGN OF THE STUDENT AT-RISK PREDICTION SYSTEM

1.11 System Architecture

The system can be organized into five layers. The data layer stores student academic and engagement records. The preprocessing layer checks data quality and converts raw values into a consistent format. The model layer learns patterns from historical examples. The prediction layer produces an AtRisk

classification for a new record. The presentation layer displays the result and relevant explanation to authorized faculty members. Separating these layers makes the system easier to test, maintain and update.

1.12 Functional Requirements

The application should accept student records, validate required fields, preprocess data, train a classifier, predict risk categories, calculate evaluation metrics and display results. It should also allow authorized users to review the record used for a prediction and record whether an intervention was provided.

1.13 Non-Functional Requirements

The system should be reliable, secure, understandable, maintainable and responsive. It should protect student information through authentication and access control. Predictions should be reproducible using a recorded dataset version, model version and preprocessing configuration. The interface should use clear language and avoid presenting a prediction as a guaranteed outcome.

1.14 Example of a Prediction Record

A sample record may contain Attendance = Low, Assignments = Irregular, LMS = Low, PriorCGPA = Weak and Mentoring = No. The trained classifier may predict AtRisk = Yes. This output should be treated as a signal for a supportive review, such as checking attendance barriers or offering academic guidance, rather than as a final judgement about the student.

TASK 2 MODEL SELECTION AND COMPARISON

2.9 Candidate Algorithms

A Decision Tree is suitable when interpretability is important because its decisions can be represented as IF–THEN rules. Logistic Regression provides a probability estimate and a relatively simple baseline. Random Forest combines multiple trees and may improve robustness, although its explanations are less direct. K-Nearest Neighbors classifies a record using nearby examples, but it can be sensitive to feature scaling and the choice of distance measure.

2.10 Model Selection Procedure

First, establish a simple baseline. Next, train candidate algorithms using the same training split and preprocessing pipeline. Compare recall, precision, F1-score and accuracy on validation data. Select a model that balances detection of at-risk students with the burden of unnecessary alerts. Finally, evaluate the chosen model once on the held-out test set.

2.11 Why Accuracy Alone Is Not Enough

If most students are not at risk, a classifier that predicts No for every record may obtain high accuracy while failing to identify students who need support. Recall and the confusion matrix are therefore necessary. Precision is also useful because excessive false alerts can overwhelm faculty and reduce trust in the system.

TASK 3 WORKED CONCEPT LEARNING EXAMPLE

3.6 Illustrative Hypothesis Representation

Consider the hypothesis <Attendance = Low, Assignments = Irregular, LMS = *, PriorCGPA = Weak>. The wildcard symbol * means that any value of LMS is accepted. This hypothesis covers only examples whose attendance, assignment status and prior CGPA match the stated conditions, regardless of LMS activity.

3.7 Coverage Test

To test whether an instance is covered, compare each non-wildcard condition with the corresponding feature value in the instance. If all required values match, the instance is covered and classified as positive by the hypothesis. If any required value differs, the instance is not covered. This simple test is the basis for updating hypotheses during concept learning.

3.8 Concept Learning Assumptions

The feature set is assumed to contain enough information to describe the target concept, the examples are assumed to be labelled, and the learner searches only within the chosen hypothesis space. If the true concept cannot be represented by the hypothesis language, the learner may not find a perfectly consistent solution even when the data labels are correct.

TASK 4 CANDIDATE ELIMINATION ALGORITHM AND TRACE INTERPRETATION

4.4 Candidate Elimination Pseudocode

Initialize S to the most specific hypothesis and G to the set of maximally general hypotheses. For each training example: if the example is positive, remove from G any hypothesis that fails to cover it; generalize members of S minimally when required; remove hypotheses from S that are more general than other members. If the example is negative, remove from S hypotheses that cover it; specialize members of G minimally when required; remove hypotheses from G that are less general than other members. At the end, return S and G as the boundaries of the Version Space.

4.5 How to Read a Boundary Trace

A trace table should show the example number, example label, current S boundary and current G boundary. After a positive example, the S boundary generally becomes more general or remains unchanged. After a negative example, the G boundary generally becomes more specific or remains unchanged. These tendencies are useful checks when reviewing a hand-worked trace.

4.6 Important Caution

The exact boundary updates depend on the attribute domains and the permitted hypothesis representation. A valid specialization must exclude the negative instance while remaining consistent with the positive examples. If the provided data includes conflicting labels for identical feature vectors, the Version Space can become empty.

TASK 5 VERSION SPACE AND ACTIVE LEARNING

5.4 Version Space as Uncertainty

The learner does not necessarily know one unique target hypothesis after observing a limited dataset. Instead, it retains all hypotheses consistent with the observations. As more labelled examples arrive, the Version Space usually shrinks. This process represents learning as elimination of inconsistent possibilities.

5.5 Example Query Strategy

Suppose two remaining hypotheses predict Yes for a candidate instance and four predict No. If the true label is obtained, one of the two groups is eliminated. A query that produces a more balanced split may be more informative because either answer removes a larger portion of the current candidate set.

5.6 Benefits and Challenges

Active learning can reduce the number of examples that need to be labelled by selecting informative queries. However, it requires access to an oracle or expert who can provide correct labels. Query selection may also be computationally expensive when the hypothesis space is very large.

TASK 6 ID3 WORKED CALCULATION AND TREE INTERPRETATION

6.5 Entropy Calculation for a Balanced Dataset

For a dataset containing 7 Yes and 7 No examples, $p(\text{Yes}) = 7/14 = 0.5$ and $p(\text{No}) = 7/14 = 0.5$. Entropy is $-0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 0.5(1) + 0.5(1) = 1$ bit. This indicates maximum impurity for a binary classification dataset.

6.6 Interpreting Information Gain

The information gain values calculated for Dataset B are Attendance = 0.1636, Assignments = 0.0611, LMS = 0.1369 and PriorCGPA = 0.4083. PriorCGPA has the greatest value, so ID3 selects it as the root attribute. It gives the largest immediate reduction in uncertainty among the attributes considered.

6.7 Example IF THEN Rules

A decision tree can be translated into rules. For example: IF PriorCGPA = Weak AND Attendance = Low, THEN AtRisk = Yes. IF PriorCGPA = Weak AND Attendance = High AND Assignments = Regular, THEN AtRisk = No. Rules make the tree's reasoning easier to communicate, though they must follow the exact branches of the constructed tree.

TASK 7 GREEDY SEARCH: STRENGTHS AND WEAKNESSES

7.4 Strengths of Greedy Search

Greedy search is straightforward, computationally practical and interpretable. It makes a local decision using an explicit criterion—in ID3, information gain. The method is easy to implement and can produce useful decision rules for small categorical datasets.

7.5 Weaknesses of Greedy Search

A locally best split may not lead to the smallest or most accurate complete tree. Greedy search may be sensitive to small changes in training data and can overfit when branches become too specific. It does not generally backtrack to reconsider earlier attribute choices.

7.6 Possible Improvements

Tree depth limits, minimum samples per leaf, pruning, cross-validation and alternative split criteria can be used to improve generalization. These settings should be selected using validation data rather than the final test set.

TASK 8 DETAILED EVALUATION AND ERROR ANALYSIS

8.5 Illustrative Confusion Matrix Layout

The confusion matrix can be represented as a 2 × 2 table with actual labels as rows and predicted labels as columns. The four cells are: Actual Yes / Predicted Yes = TP; Actual Yes / Predicted No = FN; Actual No / Predicted Yes = FP; Actual No / Predicted No = TN. Count each test prediction into one cell before calculating performance measures.

8.6 Error Analysis

False negatives are students who are actually at risk but are predicted as not at risk. False positives are students predicted as at risk who are not labelled at risk in the test data. Reviewing these records may reveal missing attributes, inconsistent labels, weak model boundaries or data-quality problems.

8.7 Threshold and Intervention Trade-Off

When a model outputs a probability, the decision threshold can be adjusted. A lower threshold usually flags more students and may increase recall, but it can also increase false positives. A higher threshold may reduce alerts but miss more at-risk students. The threshold should be selected with educators and consider the available support resources.

TASK 9 IMPLEMENTATION, TESTING AND FUTURE SCOPE

9.5 Testing Plan

Unit testing should verify data validation, entropy calculation, information gain, prediction and metric calculation separately. Integration testing should verify that preprocessing and model prediction work together. System testing should verify the complete workflow from input records to report display. Testing should include valid records, missing values, unknown category values and contradictory labels.

9.6 Sample Test Cases

Test Case 1: A valid student record is submitted; expected result is a valid prediction. Test Case 2: A required feature is missing; expected result is a validation message or documented missing-value handling. Test Case 3: An unknown categorical value is provided; expected result is safe handling without a crash. Test Case 4: The model is evaluated on held-out data; expected result is a confusion matrix and evaluation metrics. Test Case 5: An unauthorized user tries to access student records; expected result is access denied.

9.7 Future Scope

The project can be extended with a web dashboard, semester-wise risk trends, faculty intervention tracking, notifications, model comparison, explainable AI, fairness assessment and secure integration with a learning management system. Future deployment should use consent and institutional data-governance approval.

TASK 10 – EXPERIMENTAL ANALYSIS, THEORY VS EXPERIMENT, AND FINAL RECOMMENDATION

10A. Experimental Analysis (2 Marks)

10A.1 Dataset Characteristics

The early-warning system uses categorical attributes to classify students as AtRisk = Yes or AtRisk = No. The instancespace coverage is calculated as the number of unique observed attribute combinations divided by the total possible combinations, multiplied by 100.

Dataset	**Examples**	**Attributes**	**Class Balance (Yes / No)**	**Instance-Space Coverage**
Dataset A	5	6	3 / 2	5 / 96 = 5.2083%
Dataset B	14	4	7 / 7	14 / 24 = 58.3333%
Test Set T	6	4	3 / 3	6 / 24 = 25%
B + T	20	4	10 / 10	20 / 24 = 83.3333%
Dataset C	500 or more	4	Depends on generator	Depends on generated combinations

Dataset A has six attributes with cardinalities 3, 2, 2, 2, 2, and 2.

Instance space of Dataset A: $|X_A| = 3 \times 2 \times 2 \times 2 \times 2 \times 2 = 96$.

Dataset B has four attributes with cardinalities 3, 2, 2, and 2.

Instance space of Dataset B: $|X_B| = 3 \times 2 \times 2 \times 2 = 24$.

Dataset C should contain at least 500 records. Its coverage and class balance must be calculated from the actual dataset used.

10A.2 Decision Tree Performance

The decision tree is evaluated using Test Set T and 5-fold cross-validation on the combined 20 labelled records (B + T). AtRisk = Yes is treated as the positive class.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

F1-score = 2 × Precision × Recall / (Precision + Recall)

Actual / Predicted	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

TP: At-risk students correctly identified.

TN: Non-at-risk students correctly identified.

FP: Non-at-risk students incorrectly flagged.

FN: At-risk students incorrectly missed.

Recall is especially important in an academic early-warning system because false negatives represent students who need support but are not identified.

Evaluation	Accuracy	Precision	Recall	F1-score	Confusion Matrix (TP, FP, FN, TN)
Test Set T	To be measured	To be measured	To be measured	To be measured	To be measured
5-fold CV on B + T	To be measured	To be measured	To be measured	To be measured	To be measured

The exact metrics must be calculated from the predictions produced by the implementation. They depend on the treeinduction method, attribute-selection measure, tie-breaking rule, and cross-validation split.

10A.3 Effect of Training-Set Size

A learning curve is used to investigate how model performance changes as the number of training examples increases.

Training Size	Training Accuracy	Validation Accuracy	Expected Behaviour
4	To be measured	To be measured	High instability; insufficient examples
6	To be measured	To be measured	Model begins learning patterns
8	To be measured	To be measured	Improved pattern identification

10	To be measured	To be measured	Generalization may improve
12	To be measured	To be measured	Validation performance may stabilize
14	To be measured	To be measured	More reliable learning
Larger sizes using Dataset C	To be measured	To be measured	Larger data supports evaluation

Graph: Training size should be placed on the X-axis and accuracy on the Y-axis. Training accuracy and validation accuracy should be plotted as separate lines.

Analysis: A small training set may produce a tree that memorizes individual examples. Increasing the training size provides more examples of different student conditions and may improve validation accuracy. However, increased data does not guarantee better performance if labels are noisy or features are weak.

10A.4 Effect of Model Complexity

Maximum Depth	**Training Accuracy**	**Validation Accuracy**	**Nodes**	**Leaves**
0 (root only)	To be measured	To be measured	1	1
1	To be measured	To be measured	To be measured	To be measured
2	To be measured	To be measured	To be measured	To be measured
3	To be measured	To be measured	To be measured	To be measured
4	To be measured	To be measured	To be measured	To be measured
Full depth	To be measured	To be measured	To be measured	To be measured

A shallow tree may underfit because it cannot represent complex patterns.

A fully grown tree may overfit by learning noise and individual training records.

A controlled tree can improve generalization by limiting depth or requiring a minimum number of examples at a leaf.

The best depth should be selected using validation accuracy and recall rather than training accuracy alone.

Graph: Maximum depth on the X-axis; training and validation accuracy on the Y-axis. Report the number of nodes and leaves at every depth.

10A.5 Effect of Label Noise

Dataset C is used to investigate the effect of incorrect labels. Label noise is injected by randomly changing selected class labels. The same data partitions should be retained at each noise level.

Noise Rate	**Fully Grown Accuracy**	**Fully Grown Nodes**	**Controlled Tree Accuracy**	**Controlled Tree Nodes**
0%	To be measured	To be measured	To be measured	To be measured
5%	To be measured	To be measured	To be measured	To be measured
10%	To be measured	To be measured	To be measured	To be measured

Analysis: A fully grown decision tree may create additional branches to classify inconsistent records. This can increase tree size and reduce generalization accuracy. A controlled or pruned tree avoids creating branches for every noisy record and may improve validation performance and interpretability.

10A.6 Candidate Elimination vs Decision Tree

Feature	**Candidate Elimination**	**Decision Tree**
Hypothesis representation	Conjunctive hypotheses	Hierarchical decision rules
Learning method	Version-space boundary updates	Heuristic recursive splitting
Disjunction	Limited in standard representation	Can represent disjunction through branches
Noise handling	Highly sensitive to inconsistent labels	Can tolerate some noise through pruning/controlled growth
Unseen combinations	Depends on surviving hypotheses	Uses branch rules and fallback policies
Interpretability	General and specific boundaries	Human-readable IF–THEN rules
Complexity	Version-space boundaries	Nodes and leaves

Suitability for large data	Limited in basic form	More suitable with controlled learning

Candidate Elimination is expected to collapse when contradictory examples cannot be represented by a single consistent hypothesis. A decision tree may still return a prediction because it selects a hypothesis using attribute-selection heuristics.

10B. Theory vs Experimental Behaviour (2 Marks)

10B.1 Theoretical Expectations and Experimental Behaviour

Theoretical Expectation	**Experimental Behaviour**	**Interpretation**
A fully grown tree may overfit	Training accuracy may be high while validation accuracy is lower	Supports overfitting theory
Candidate Elimination requires consistent examples	Contradictory labels may make the version space empty	Supports consistency requirement
More examples improve learning	Validation performance may improve with training size	More data provides broader evidence
Conjunctive hypotheses have limited representation	Candidate Elimination may fail on disjunctive concepts	Representation restricts learning
Decision trees represent complex concepts	Branches capture different risk conditions	More flexible representation
Information Gain reduces entropy	Root selection depends on entropy reduction	Greedy heuristic affects structure
Gain Ratio reduces preference for many-valued attributes	Root or branch selection may change	Selection measure influences tree
Increasing tree depth increases complexity	Nodes and leaves generally increase	Greater complexity may overfit

10B.2 Dataset Characteristics

Dataset A contains only five examples and has low instance-space coverage. Candidate Elimination therefore constructs boundaries from limited evidence. Dataset B contains 14 training examples with equal class balance but does not cover every possible combination. Dataset C contains at least 500 examples and provides a larger basis for evaluating learning

curves, tree complexity, and noise handling. Its usefulness depends on whether its generation rule represents realistic student behaviour.

10B.3 Inductive Bias

Candidate Elimination assumes that the target concept can be represented using its hypothesis language. Its standard conjunctive representation restricts the patterns it can learn. Decision trees represent knowledge through attribute tests and branches, allowing them to model combinations and disjunctions more naturally.

For example, Candidate Elimination may represent Attendance = Low AND Assignments = Irregular. A decision tree can represent separate risk conditions through branches, such as Attendance = Low OR PriorCGPA = Weak. **10B.4**

Information Gain and Gain Ratio

Information Gain measures the reduction in entropy after splitting on an attribute.

$IG(S, A) = H(S) - \Sigma_v (|S_v| / |S|) H(S_v)$

Gain Ratio = IG(A) / SplitInformation(A)

Information Gain may prefer attributes that produce many branches. Gain Ratio considers the distribution of examples among branches and reduces this preference. The heuristic can affect root attribute, depth, leaves, and predictive performance.

10B.5 Limitations and Threats to Validity

Dataset A and Dataset B are small and may not represent real student populations.

Results depend on the selected training and test examples.

Synthetic Dataset C may not represent actual academic risk.

Random label noise may not reflect real advisor mistakes.

Decision trees can be sensitive to small training-data changes.

Cross-validation on only 20 examples may produce unstable estimates.

Accuracy alone may hide at-risk students who are missed.

Tie-breaking rules may produce different tree structures.

The experiments provide useful evidence but do not guarantee real-world performance.

10C. Evidence-Based Recommendation (2 Marks)

Recommended Approach: Controlled Decision Tree with Human-Readable Explanations

Evaluation Factor	Candidate Elimination	Controlled Decision Tree
Accuracy	Limited by representation and consistency	Can be tuned and evaluated on validation data
Recall	May fail when version space collapses	Can be adjusted using class weights and threshold
Interpretability	Hypothesis boundaries	Clear IF–THEN decision rules
Noise handling	Sensitive to contradictions	Pruning and minimum leaf size can reduce overfitting
Unseen values	Limited by hypothesis representation	Can use retraining and fallback policy
Complexity	Maintains S and G boundaries	Controlled by depth and leaf constraints
Data requirements	Consistent examples needed	Can use larger datasets and validation
Explanation	General and specific hypotheses	Decision path through the tree

Candidate Elimination is useful for understanding concept learning and version-space behaviour in small, consistent datasets. However, its strict consistency assumption makes it unsuitable as the primary model when labels are inconsistent and new attribute values appear.

A controlled decision tree is more appropriate because it can learn interpretable rules, handle complex patterns, and be adjusted to prioritize recall. The final model should be selected only after experimental evidence confirms its accuracy and recall. The model should minimize false negatives to satisfy the academic early-warning requirement.

TASK 11 – DAY-4 DESIGN CHALLENGE: RE-EVALUATE AND ADAPT THE DESIGN

11A. Impact Analysis (2 Marks)

11A.1 Change-versus-Assumption Table

Changed Requirement	Original Assumption	Category	Consequence

2,000 labelled records	Small dataset	Data	Larger-scale learning required
New records arrive weekly	Static dataset	Data / Algorithm	Periodic or incremental retraining
5–10% inconsistent labels	Noise-free labels	Data	Strict Candidate Elimination may fail
Attendance = Very Low	Fixed attribute values	Hypothesis space	New values require handling
New attribute combinations	Limited instance-space coverage	Hypothesis space	Model must generalize
At least 90% recall	Accuracy-focused evaluation	Evaluation	False negatives must be reduced
Human-readable explanations	Basic prediction output	Evaluation / System design	Each flagged student needs a reason

11A.2 Why E6 and N2 Are Treated Differently

E6 in Dataset A is Low, Irregular, Low, Weak, Passive, No → AtRisk = No. This is identical to E1 in all six input attributes, but E1 is labelled AtRisk = Yes. E6 therefore contradicts E1.

Candidate Elimination assumes that examples are correctly labelled and that a single consistent target concept exists. When E6 is added, the version space becomes empty because the same input combination has two different class labels. A noise-tolerant method may treat one label as inconsistent and learn from majority, confidence, or regularized evidence.

N2 in Dataset B is High, Regular, Active, Strong → AtRisk = Yes. This conflicts with original examples associating the same combination with AtRisk = No. Candidate Elimination restricted to the same attributes and a consistent conjunctive representation may fail to retain a valid hypothesis. A decision tree can still produce a classification, but a fully grown tree may create an extra branch to memorize N2. A controlled tree may treat N2 as noisy evidence and avoid excessive branching.

Conclusion: E6 and N2 are contradictory examples. Candidate Elimination reports a consistency failure, while a noisetolerant decision tree can still return a prediction.

11B. Re-evaluation of the Original Design (2 Marks)

11B.1 Candidate Elimination with E6, E7, and E8

The original Dataset A contains E1–E5. Candidate Elimination maintains S, the maximally specific consistent hypotheses; G, the maximally general consistent hypotheses; and the version space, all hypotheses consistent with the observed examples.

Stage	Example	Result
Initial	E1–E5	Non-empty version space
Added independently	E6	Contradiction with E1
S boundary after E6	E6	No consistent S boundary
G boundary after E6	E6	No consistent G boundary
Version Space after E6	E6	Empty

Experiment 2: Add E7 and then E8 cumulatively after E1–E5.

Stage	Example Added	Result
Initial	E1–E5	Version space exists
Step 1	E7	Update S and G using the new positive example
Step 2	E8	Update S and G using the next positive example

E7 = Medium, Regular, High, Weak, Passive, No → Yes. E8 = Low, Regular, High, Strong, Active, Yes → Yes. These positive examples introduce different attribute values and increase the need for generalization. The conjunctive hypothesis must cover the positive examples while excluding the negative examples. If this is not possible in the hypothesis language, the version space collapses.

Execution logs should print S, G, and version-space status after each example. A collapsed version space should be reported explicitly.

11B.2 Decision Tree After Adding N1–N3

Record	Attendance	Assignments	LMS	PriorCGPA	AtRisk
N1	Very Low	Irregular	Passive	Weak	Yes

N2	High	Regular	Active	Strong	Yes
N3	Low	Regular	Passive	Weak	Yes

Dataset	**Examples**	**AtRisk = Yes**	**AtRisk = No**
Original B	14	7	7
Added N1–N3	3	3	0
Combined	17	10	7

The new records may change the root attribute because information gain changes. A branch may be created for Attendance = Very Low. The High, Regular, Active, Strong branch becomes inconsistent because N2 has a positive label. N3 may influence the Low attendance branch. Tree depth and leaf count may increase if the tree is fully grown.

Handling N1: Attendance = Very Low

If the tree is retrained with N1, create a branch for Very Low if the split criterion supports it.

If an unseen value occurs during deployment, use a documented fallback.

The fallback may use the parent node majority class or a calibrated probability.

Flag records for human review when prediction confidence is low.

Candidate Elimination is not guaranteed to meet 90% recall or the explanation requirement. A controlled decision tree can provide human-readable paths and can be tuned for recall, but 90% recall must be verified experimentally.

11C. Adaptation or Redesign (2 Marks)

11C.1 Final Design Decision: REDESIGN

The original design is redesigned into a noise-tolerant, recall-focused decision-tree early-warning system.

11C.2 Revised Learning Method

Component	**Original Design**	**Revised Design**
Hypothesis representation	Conjunctive Candidate Elimination	Decision tree with multiple branches
Learning algorithm	Candidate Elimination and ID3	Controlled ID3 or pruned decision tree

Noise handling	Assumes consistent labels	Pruning, minimum leaf size, label audit
Attribute selection	Information Gain	Information Gain or Gain Ratio
New attribute values	Fixed domains	Unknown-value and fallback policy
Recall	Accuracy-based evaluation	Recall-focused tuning
Explanation	General hypothesis	IF–THEN decision path
Data updates	Static	Weekly retraining and monitoring

11C.3 Revised Algorithm Outline

1. Collect student records.
2. Audit labels and handle missing values.
3. Encode categorical attributes.
4. Split data into training, validation, and test sets.
5. Train a controlled decision tree.
6. Apply class weights to give more importance to AtRisk = Yes.
7. Tune the decision threshold using validation data.
8. Calculate accuracy, precision, recall, and F1-score.
9. If recall < 90%, adjust class weights or threshold, then retrain and validate.
10. If recall ≥ 90%, evaluate the final model on the test set.
11. Generate an IF–THEN explanation for every flagged student.
12. Send the explanation to an academic advisor.
13. Monitor performance weekly.
14. Retrain when new records become available. **11C.4 Recall and Precision Trade-off**

Recall = TP / (TP + FN). To achieve at least 90% recall, the model must identify at least 90% of actual at-risk students. Class weighting and threshold adjustment can increase recall; however, this may increase false positives and reduce precision. The threshold must therefore be selected using validation results and the university's support capacity.

11C.5 Human-Readable Explanations Example Student Risk Explanation:

Attendance: Low

Assignments: Irregular

Prior CGPA: Weak

Predicted risk: AtRisk = Yes

Explanation: The student has low attendance, irregular assignment submission, and a weak prior CGPA. These conditions follow a high-risk decision path in the trained decision tree. The student is recommended for academic counselling and remedial support.

The explanation should show the actual decision path used by the model. It should not claim that an attribute causes academic failure.

11D. Evidence-Based Defence (2 Marks)

11D.1 Justification Using Learning Theory

Candidate Elimination maintains a version space of hypotheses consistent with training examples. Its basic assumption is that observed labels are correct. The new dataset contains inconsistent labels, violating this assumption. A strict Candidate Elimination implementation may produce an empty version space.

Decision-tree learning uses heuristic search to select attributes that reduce uncertainty. It can represent complex combinations through branches and is more flexible for academic risk identification.

11D.2 Justification Using Experimental Evidence

Evaluation	**Required Evidence**
Recall	At least 90% on validation/test evaluation
Precision	Report the number of false positives
Accuracy	Compare original and redesigned tree
Complexity	Compare nodes, leaves, and depth

Noise	Evaluate performance under 5% and 10% label noise
Interpretability	Show example decision paths
Generalization	Use cross-validation and held-out testing

The redesign is justified if the controlled decision tree achieves the required recall while maintaining acceptable precision, complexity, and interpretability.

11D.3 Remaining Limitations Advisor

labels may remain inconsistent.

New attribute values may require retraining.

A high recall requirement can increase false-positive predictions.

A decision tree may change structure when new records are added.

Academic risk may depend on factors not present in the dataset.

11D.4 Monitoring Plan

Monitor recall of AtRisk = Yes students weekly.

Monitor precision and false-positive rate.

Track the number of students flagged.

Track tree depth and number of leaves.

Track new attribute values and unseen combinations.

Compare model predictions with advisor decisions.

Monitor changes in student performance patterns.

If recall falls below 90%, the model should be reviewed, retrained, and evaluated again before continued deployment.

FINAL CONCLUSION

The experimental analysis demonstrates the importance of selecting an appropriate learning method based on dataset characteristics, hypothesis representation, model complexity, and evaluation requirements. Candidate Elimination is useful for studying concept learning and version-space behaviour, but its strict consistency assumption limits its suitability when student labels contain noise. Decision-tree learning provides a more flexible and interpretable

representation through hierarchical decision rules. A controlled or pruned decision tree can reduce overfitting and provide human-readable explanations for academic advisors. Under the updated requirements, the system should be redesigned to use a recall-focused decision tree with class weighting, validation-based threshold selection, explicit handling of unseen attribute values, and weekly performance monitoring. The model should be deployed only after experimental testing confirms that it achieves at least 90% recall on at-risk students while maintaining acceptable precision and interpretability. Human review remains essential because the system is intended to support academic decisions rather than replace advisors.